

- 1 (a) (i) number of molecules B1 [1]
- (ii) mean square speed B1 [1]
- (b) (i) 1. $pV = nRT$ C1
 $n = (6.1 \times 10^5 \times 2.1 \times 10^4 \times 10^{-6}) / (8.31 \times 285)$ C1
 $n = 5.4 \text{ mol}$ A1 [3]
2. *either* $N = nN_A$ C1
 $= 5.4 \times 6.02 \times 10^{23}$ C1
 $= 3.26 \times 10^{24}$ A1
or
 $pV = NkT$
 $N = (6.1 \times 10^5 \times 2.1 \times 10^4 \times 10^{-6}) / (1.38 \times 10^{-23} \times 285)$ (C1)
 $N = 3.26 \times 10^{24}$ (A1) [2]
- (ii) *either* $6.1 \times 10^5 \times 2.1 \times 10^{-2} = \frac{1}{3} \times 3.25 \times 10^{24} \times 4 \times 1.66 \times 10^{-27} \times \langle c^2 \rangle$ C1
 $\langle c^2 \rangle = 1.78 \times 10^6$ C1
 $c_{\text{RMS}} = 1.33 \times 10^3 \text{ m s}^{-1}$ A1
or
 $\frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times \langle c^2 \rangle = \frac{3}{2} \times 1.38 \times 10^{-23} \times 285$ (C1)
 $\langle c^2 \rangle = 1.78 \times 10^6$ (C1)
 $c_{\text{RMS}} = 1.33 \times 10^3 \text{ m s}^{-1}$ (A1) [3]
- 2 (a) (i) 1. 0.1 s, 0.3 s, 0.5 s, etc (any two) A1 [1]